# Bulk Deformation Processes - Forging

Bulk deformation processes refer to deforming the solid at stresses above the yield strength.

For this chapter, we focus on forging, the process of using compressive forces to shape metal.

Simple forging is the easiest kind, involving simple hammering on an anvil, however more typically dies are used to compress the part. These compressive forces are either applied gradually or by impact.

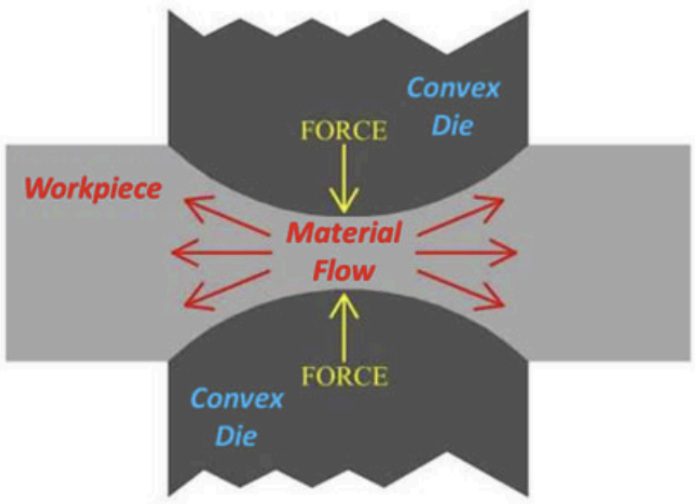
Generally, forging alters the microstructure of the metal and seals cracks, improving strength, hence this process is useful for high strength components like crankshafts, gears, etc.

Effects by Forging

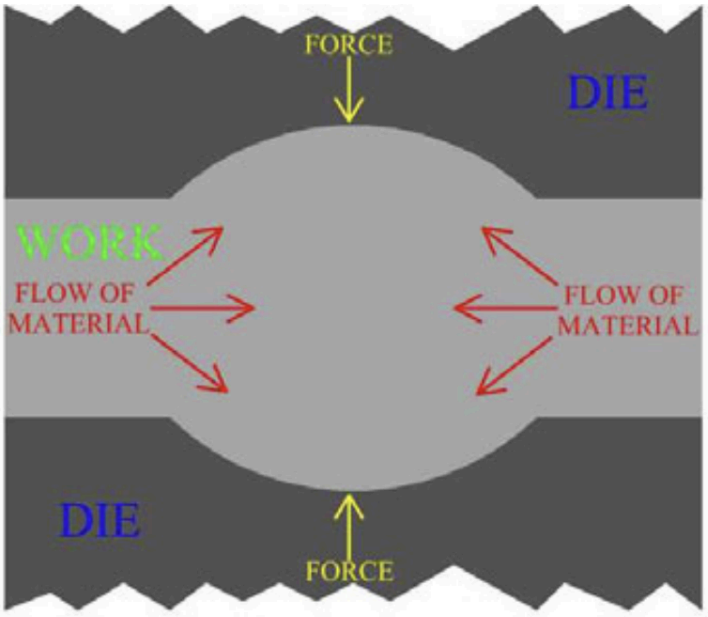
- Microstructure is aligned along contour of part, instead of machining (uniform direction) or casting (no direction), typically along direction needing high strength, allowing for high impact and fatigue resistance, unlike machining (more sensitive to stress corrosion).
    - microstructure flows to form fiborous microstructure, giving better mechanical properties
    - recrystallisation may be done afterwards to produce finer grains, giving better toughness and other properties.

Forging Operations

- Fullering
    - convex dies distribute metal away from fuller center for further operations, causing a reduction in cross sectional area and outward metal flow




Edging



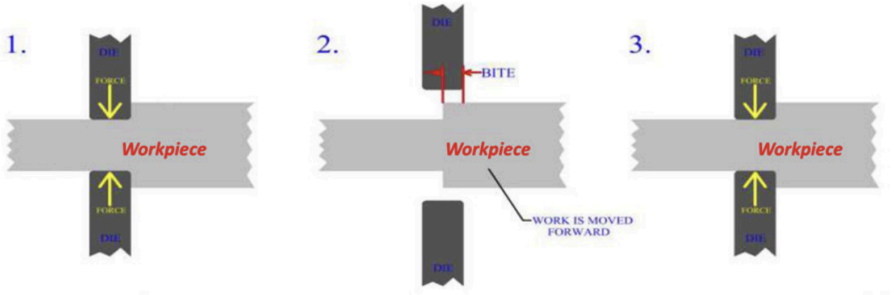

opposite of fullering, concave dies distribute material into the middle for further operations, metal flow is inward but free to move laterally.

Cogging/Drawing

process thins material while increasing length, by compressing the material at sections of the material along its length.

the amount the die travels along the material length is called the bite (approx 40 to 75 % of the die width)

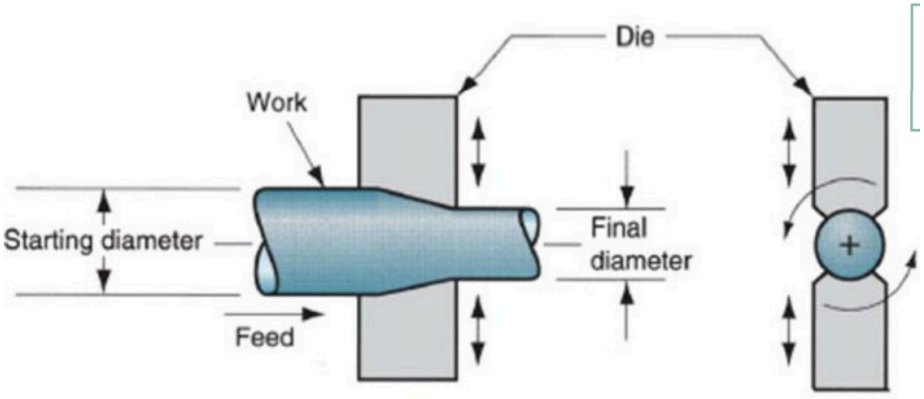
able to lengthen parts even with small machinery with lesser forces




Swaging

using two circular concave dies, metal is shaped into circular profiles of smaller diameter. Multiple hammer blows are applied while the workpiece rotates in the die to maintain a circular profile.

smaller diameter profile is used for tapping, threading, or other operations.




Forging Equipment

Generally, there are two types of forging equipment, forging hammers, which apply impact to the metal, and forging presses, applying gradual force.

Forging Hammers

Board Hammer

friction rolls move the upper die and ram upward to then be released on the lower die and anvil

max energy supplied limited by height of ram (potential energy = mgh)

- Power Hammer
  - can supply greater force and energy via steam/air pressure to raise the ram, and to accelerate the ram on the downstroke alongside gravity
  - Total energy supplied

$$W = \frac{1}{2}mv^2 + pAH = (mg + pA)H$$

- Forging Presses
  - Mechanical Press
    - ram stroke is typically controlled by rotary crank, giving a lesser stroke and reducing the usable die size, but with better life
    - rated by the pressure at the end of the downstroke
  - Hydraulic Press
    - hydraulic piston moves ram at a slower rate
    - provides deeper penetration with better properties
    - full press load available at any point of the stroke, hence ideal for extrusion type forging
    - slower time causes longer contact which may cause heat loss
    - more expensive than mechanical presses and hammers

Forging Processes

- there are 3 methods to apply the forging processes:
- Open Die Forging
  - metal is forged in the open, with simple dies/flat dies, allowing metal to flow laterally
  - mainly used for larger objects at small scale, to preform the workpiece for closed die forging
  - advantages:
    - simple, inexpensive, wide range of sizes, produces good strength
  - disadvantages:
    - limited shape complexity, difficult to hold close tolerances (further processes like machining needed)

low scale

poor utilisation of material

high skill needed

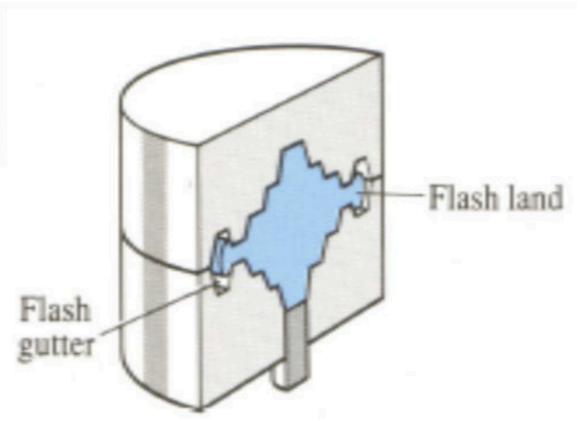
Closed Die/Impression Forging

two metal dies form a closed cavity when applying the high pressure, with the profile of the desired final shape

used for smaller components, producing close tolerances.

However, the dies required are expensive, and to achieve the final step of forging requires skill to design the intermediate steps to make the final precision part

In closed die forging, there are special spaces called flashes,

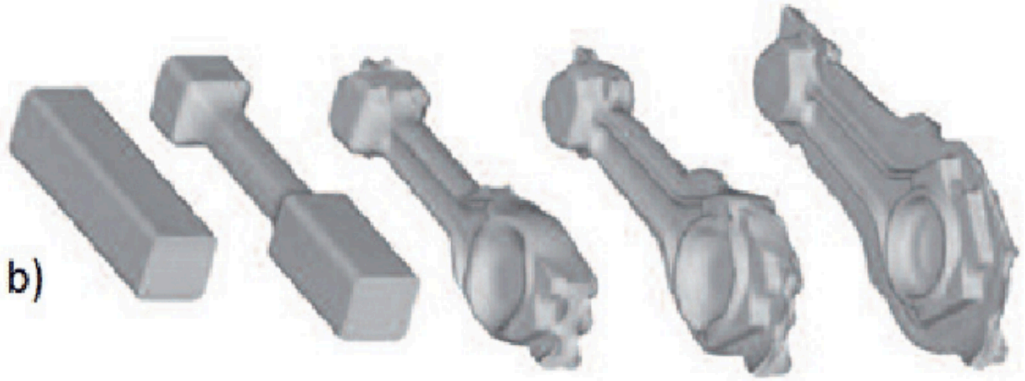



which is a safety buffer for excess material to flow into to build up high pressure, ensuring all the metal fills the whole die cavity (also prevents excessive pressures)

With open and closed die forging, the process briefly involves:

Preforming the stock with an open die arrangement

Using a block die to form the rough shape, then using one or multiple finishing dies to achieve the specific features, tolerances, and surface finish, before flashes are removed.



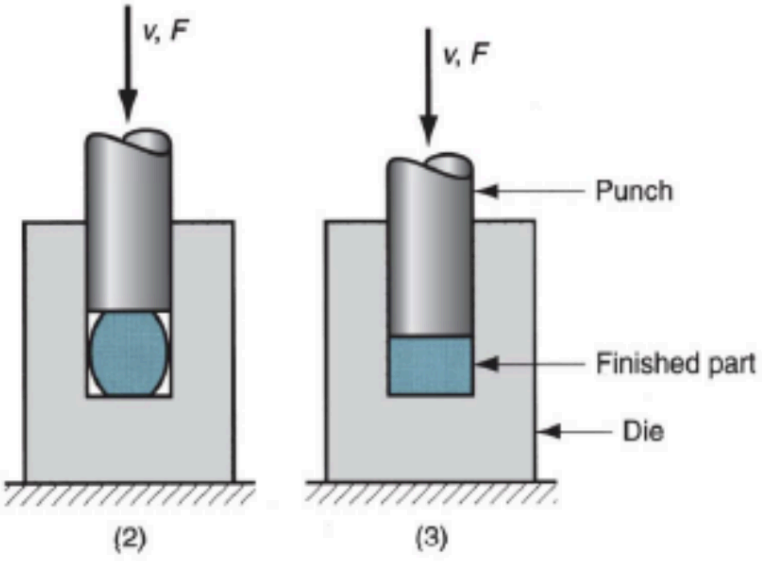

- advantages
  - less material is wasted (good utilisation of material)
  - better mechanical properties than open die forgings
  - multi part dies can be used to create more complex parts
  - higher force presses permittable
  - good dimensional accuracy
  - large scale
- disadvantages
  - high die cost
  - machining needed to remove flashes, which cause poorer dimensional accuracies at flash points.
- To ensure a part is manufacturable with closed die casting,
  - smooth metal flow should be accounted for, meaning
    - use symmetric dies instead of thin and long sections
    - avoid sudden section shape changes
    - generous filets and drafts where possible
      - 3 to 5 deg outside draft, 7 to 10 draft inside (??)
  - minimise flashes
  - if dies with inclined angles are used, counter locks should be made to prevent dies from sliding apart due to side thrust (component force pushing dies away

Flashless Forging

same as closed die, however no flash is formed, by employing net shape forming (process that forms the main shape immediately, without need for further processes)

To achieve this, proper volume control is needed of the material and the die cavity, otherwise complete filling or die jamming/failure may occur, for under or oversized workpieces respectively.

Hence, this best works for simple and symmetrical geometries, and for soft alloys like aluminium and magnesium. (lower load and temp needed, compared to steel and titanium)




Forging Defects

for ribs and webs, ribs are sections perpendicular to the parting line, while webs are parallel to the parting line.

thin ribs are harder to fill and need more force, hence thicker ribs should be designed.

likewise, thin webs are hard to fill but also cool faster which may cause warping or tears.

surface may crack during forging process, if it is not malleable enough.

if the work is undersized, laps occur, e.g. when material in a web buckles during forging and creates divots called laps. Hence, web thickness should

be increased.

However, too thick of webs may cause material to flow into already forged portions, causing internal cracks